



DENDRITIC CELLS

LXR signaling controls homeostatic dendritic cell maturation

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Dendritic cells (DCs) mature in an immunogenic or tolerogenic manner depending on the context in which an antigen is perceived, preserving the balance between immunity and tolerance. Whereas the pathways driving immunogenic maturation in response to infectious insults are well-characterized, the signals that drive tolerogenic maturation during homeostasis are still poorly understood. We found that the engulfment of apoptotic cells triggered homeostatic maturation of type 1 conventional DCs (cDC1s) within the spleen. This maturation process could be mimicked by engulfment of empty, nonadjuvanted lipid nanoparticles (LNPs), was marked by intracellular accumulation of cholesterol, and was highly specific to cDC1s. Engulfment of either apoptotic cells or cholesterol-rich LNPs led to the activation of the liver X receptor (LXR) pathway, which promotes the efflux of cellular cholesterol, and repressed genes associated with immunogenic maturation. In contrast, simultaneous engagement of TLR3 to mimic viral infection via administration of poly(I:C)-adjuvanted LNPs repressed the LXR pathway, thus delaying cellular cholesterol efflux and inducing genes that promote T cell-mediated immunity. These data demonstrate that conserved cellular cholesterol efflux pathways are differentially regulated in tolerogenic versus immunogenic cDC1s and suggest that administration of nonadjuvanted cholesterol-rich LNPs may be an approach for inducing tolerogenic DC maturation.

INTRODUCTION

Dendritic cells (DCs) are professional antigen-presenting cells that bridge innate and adaptive immunity by presenting antigens to T lymphocytes in lymphoid tissues (1–4). Precursor conventional DCs (pre-cDCs) develop in the bone marrow (5, 6) yet rapidly seed tissues while undergoing a differentiation process either into XCR1-expressing type 1 conventional DCs (cDC1s), specialized in cross-presenting antigen to CD8 T cells, or into SIRP1α-expressing type 2 conventional DCs (cDC2s), specialized in presenting exogenous antigens to CD4 T cells (1, 4, 7–9). In peripheral tissues, cDCs are in an immature or resting state, poised to recognize incoming pathogens via pattern recognition receptors (PRRs) (1). Maturation of DCs, induced by triggering of germline-encoded PRRs, is a crucial determinant of T cell activation and forms the basis of how the discrimination between self and non-self is

regulated (10). After uptake and recognition of microbial antigens, DCs undergo a coordinated maturation process to become immunogenic, migrate to T cell-rich areas, and have the capacity to stimulate naïve T cells. The ability to activate T cells is acquired through the digestion of antigen and presentation on MHC molecules (signal 1 for T cell activation), the progressive expression of costimulatory molecules for T cell activation (signal 2), and the secretion of T helper-polarizing cytokines (signal 3) (1, 11).

However, even in steady state, many DCs already express costimulatory molecules in the apparent absence of infectious or damage-associated stimuli (1). This form of homeostatic maturation is required for peripheral tolerance to self-antigen of autoreactive T cells that have escaped negative selection in the thymus. Autoreactive T cells that see antigen presentation by mature DCs in homeostatic conditions undergo a few rounds of T cell division

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before being deleted or develop into peripheral regulatory T cells (T_{regs}) that can suppress autoreactive responses; hence, homeostatic DC maturation has also been called tolerogenic maturation (12–19). Conceptually, homeostatic DC maturation does not differ substantially from maturation induced by microbial insults because DCs also migrate to the lymph node in a CCR7-dependent manner (20–22) and express peptide-loaded major histocompatibility complex (MHC) molecules along with several costimulatory molecules (23–25). Recent detailed gene expression studies have, however, unraveled discrete differences between immunogenic and tolerogenic DC maturation that could be used to discriminate these cells for functional analysis (26, 27).

Although it has been acknowledged for many years that DC maturation does not necessarily lead to induction of effector immunity (19), the factors that initiate homeostatic maturation of DCs remain poorly understood. Prior work proposed that uptake of apoptotic material by cDC1s would induce tolerance to self-antigens, but this was not connected to DC maturation, and other studies indicated that DC maturation would be prevented by apoptotic cell (AC) uptake (12, 18, 20, 28–33). Several other maturation signals have been proposed, such as triggering of PRRs by the commensal microbiome (or their derived products) and tonic cytokine receptor activation via type I interferon (IFN) or tumor necrosis factor- α production, but none of these signals were found to be essential (26, 27, 34). Therefore, it has been concluded that homeostatic maturation might be a stochastic process, resulting from a DC maturation module being turned on to benefit immune homeostasis rather than an instructive process via endogenous or exogenous stimuli (1, 26).

Here, we mapped the molecular processes driving homeostatic cDC1 maturation in the spleen, where DCs can migrate from the red pulp, where antigen uptake takes place, to the white pulp, where DCs interact with T cells (3, 29, 35–37). We focused our analysis on cDC1s that mediate tolerance or immunity to dead cell-derived antigens depending on the context in which ACs are encountered (29, 38). We found that homeostatic maturation in cDC1s was driven by AC engulfment, a process that can be modeled by uptake of nonadjuvanted lipid nanoparticles (LNPs). Moreover, we identified a role for the liver X receptor (LXR) pathway, a key intracellular sensor and regulator of cholesterol metabolism, in controlling tolerogenic versus immunogenic DC maturation among cDC1s.

RESULTS

Homeostatic mature cDC1s express canonical DC maturation markers

In steady-state conditions, about 10% of all cDC1s (and cDC2s) in the spleens of specific pathogen-free mice expressed high levels of the DC maturation markers MHCII and CCR7 (fig. S1A and gating strategy in fig. S2). Consistent with these cells having a mature state, homeostatic CCR7⁺ cDC1s showed increased expression of costimulatory molecules, such as CD86 or CD40, although not to the same extent as was observed during immunogenic maturation, which was induced by injecting mice with the Toll-like receptor 3 (TLR3) ligand, poly(I:C) (pIC; fig. S1B). Notably, the expression of the costimulatory marker CD80 in homeostatic mature cDC1s was even lower than in immature CCR7[−] cDC1s. During homeostasis, T_{regs} have been shown to reduce costimulatory molecule expression

on migratory DCs through cytotoxic T lymphocyte-associated protein 4 (CTLA4)-mediated transendocytosis (39). We found that CD80 expression levels on migratory homeostatic DCs were increased by injecting mice with a CTLA4-blocking antibody (fig. S1C), consistent with a loss of CTLA4-mediated repression of migratory DCs by T_{regs} . Housing the mice in germ-free conditions did not alter homeostatic maturation, excluding a role for commensals in driving the homeostatic maturation program (fig. S1D), as noted previously (34).

To detect gene signatures associated with homeostatic maturation, we sorted immature CCR7[−] and mature CCR7⁺ cDC1s from the spleen and analyzed their transcriptomic profile via bulk RNA sequencing. Ingenuity Pathway Analysis determined that gene expression in homeostatically matured DCs cells was enriched in "NF- κ B signaling," "DC Maturation," and "Role of Pattern Recognition Receptors" (fig. S3), pathways that would also be expected during immunogenic maturation and hence referred to as "common" maturation pathways (26, 27). At the individual gene level, well-known markers of DC maturation were expressed by CCR7⁺ cDC1s (*Cd70*, *H2-M2*, and *Cd40*), as well as genes traditionally associated with DC migration and maturation-associated cytoskeletal rearrangements (e.g., *Ccr7* and *Fscn1*) or intracellular vesicle transport (e.g., *Cd63* and *Mreg*; fig. S1E and data file S1). Several of the top differentially expressed (DE) genes appeared to be associated with the transforming growth factor- β (TGF β) pathway, such as *Scube3*, a TGF β receptor II ligand (40), or the integrin *Itgb8*, crucial for activation of TGF β in T_{regs} (41). Expression of genes such as *Ccl22*, a chemokine needed for communication of dendritic cells with CCR4⁺ T_{regs} (42), and *Aldh1a2*, which catalyzes retinoic acid synthesis, suggests that homeostatic maturation of cDC1s is coupled with maintaining and inducing peripheral T_{regs} . Overall, these data illustrate a common and distinct program of homeostatic splenic DC maturation of cDC1, in line with earlier descriptions of spontaneously migrating cDCs in other tissues (26, 43).

AC engulfment drives homeostatic maturation in cDC1s

Both in steady-state conditions and the tumor environment, the gene signature of cDC1s that have engulfed ACs closely overlaps with that of steady-state migratory DCs (44, 45). Because AC engulfment has been linked to the induction of tolerance, we asked whether the process of AC engulfment regulates homeostatic maturation. To test this, CellTrace Violet (CTV)-labeled apoptotic thymocytes were injected into mice intravenously, and the engulfment of ACs by splenic DC populations was assessed by determining the percentage of CTV⁺ DCs after 2 hours (Fig. 1A). Confirming earlier observations (46), we found that only macrophages and cDC1s had the capacity to engulf ACs, whereas cDC2s did not (Fig. 1, A and B). Next, we followed engulfment of CTV-labeled ACs in the spleen over time (Fig. 1C). At early time points, a steep rise in the numbers of engulfing macrophages could be observed, but between 2 and 4 hours after injection, the numbers dropped. At that time, the percentage of CTV⁺ cDC1s was still rising, plateauing after 4 hours and subsequently declining at 12 hours after injection. To determine whether CTV-labeled ACs were still available to be engulfed by cells after 4 hours of injection, we performed a secondary engulfment experiment, where mice were given an initial injection of CTV-labeled ACs and were subsequently injected with CellTracker Red (CTR)-labeled ACs 2 hours later. These secondary

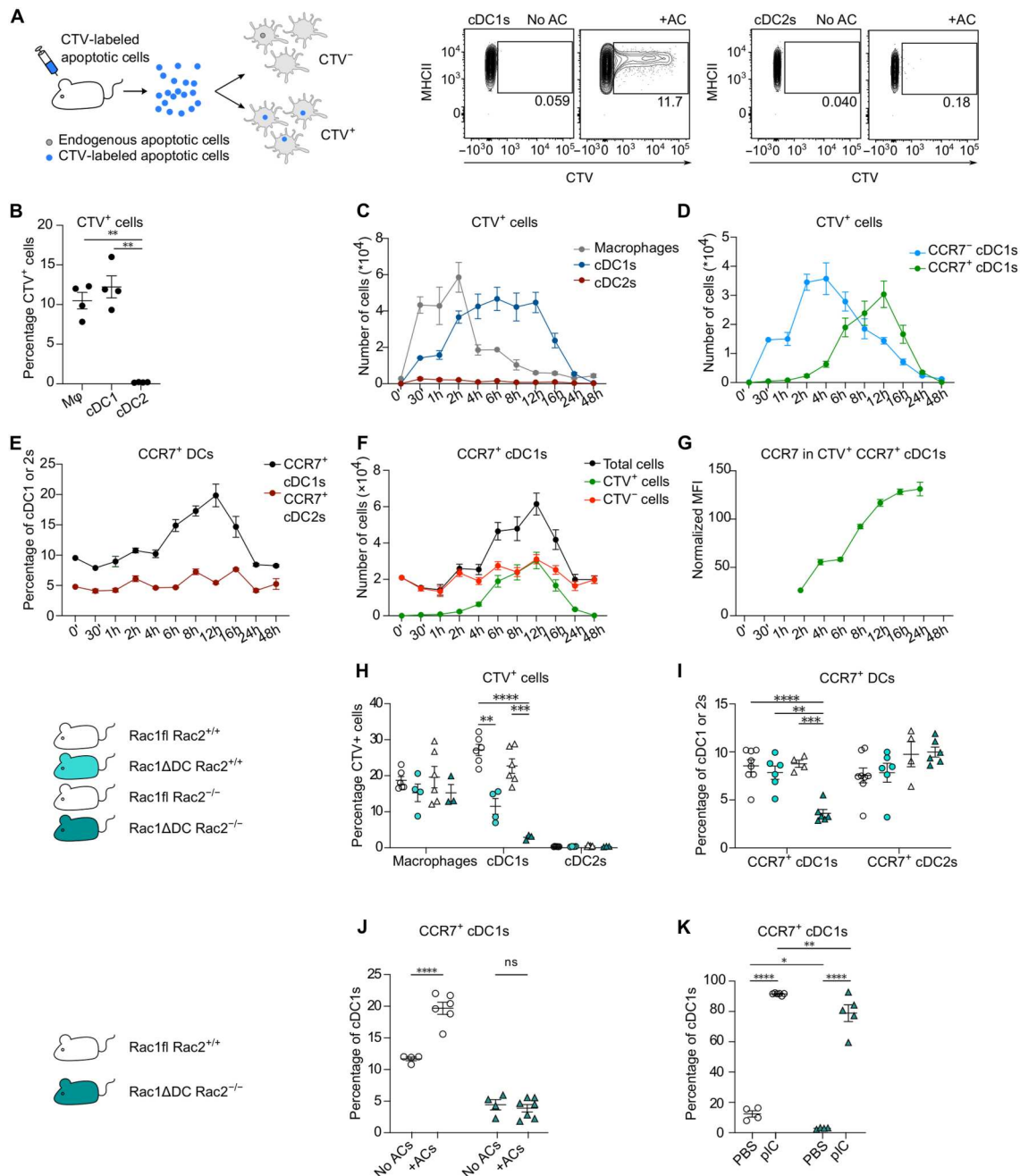


Fig. 1. AC engulfment drives homeostatic maturation in cDC1s. (A) Experimental setup. Mice were injected intravenously with CTV-labeled ACs. At dedicated time points after injection, mice were euthanized, and the uptake of ACs by DCs isolated from the spleen was measured by the presence of CTV signal. Representative FACS plots showing the engulfment in cDC1s (left) and cDC2s (right) based on CTV positivity 2 hours after injection. (B) Engulfment of CTV-labeled ACs in splenic macrophages, cDC1s, and cDC2s 2 hours after injection. One-way ANOVA, corrected for multiple testing by Tukey's multiple comparisons test with individual variances. (C and D) Kinetics of the number of CTV⁺ cells in different populations after injection of CTV-labeled ACs. (E) Kinetics of the percentage of total CCR7⁺ cDC1s and cDC2s after injection of CTV-labeled ACs. (F) Kinetics of the number of total, CTV⁺, and CTV⁻ CCR7⁺ cDC1s after injection of CTV-labeled ACs. (G) Kinetics of the normalized MFI of CCR7 in CTV⁺ CCR7⁺ cDC1s. The MFI was normalized to the MFI of CTV⁻ CCR7⁺ cDC1s. (H) Engulfment of CTV-labeled ACs in splenic macrophages, cDC1s, and cDC2s of Rac1^{fl/fl} or ΔDC Rac2^{+/-} or Rac2^{-/-} 2 hours after injection. Two-way ANOVA, corrected for multiple testing by Tukey's multiple comparisons test with individual variances. (I) The percentage of CCR7⁺ cDC1s (left) and cDC2s (right) of Rac1^{fl/fl} or ΔDC Rac2^{+/-} or Rac2^{-/-} in steady state. Two-way ANOVA, corrected for multiple testing by Tukey's multiple comparisons test with a single pooled variance. (J) The percentage of CCR7⁺ cDC1s in the spleen in Rac1^{fl/fl} Rac2^{+/-} (left) or in Rac1ΔDC Rac2^{-/-} (right) mice 12 hours after injection of CTV-labeled ACs. Two-way ANOVA corrected for multiple testing by controlling the false discovery rate. Asterisks represent the *q* value. (K) The percentage of CCR7⁺ cDC1s in the spleen of Rac1^{fl/fl} Rac2^{+/-} (left) or in Rac1ΔDC Rac2^{-/-} (right) mice 16 hours after injection of 50 μg pIC. Two-way ANOVA corrected for multiple testing by controlling the false discovery rate. Asterisks represent the *q* value. The mean ± SEM is shown in (B to K). The x axis in (C to G) indicates the time after injection of CTV-labeled ACs. *N* = 1 for the 0' time point, and *n* = 5 for all other time points in (C to G). Representative of at least two experiments in (A to K). **P* < 0.05, ***P* < 0.01, ****P* < 0.001, and *****P* < 0.0001; ns, not significant.

engulfment experiments revealed that macrophages were still able to engulf (as shown by the increase in CTR labeling) but were no longer CTV⁺, showing that all CTV-labeled cells were consumed 4 hours after the initial injection (fig. S4A). Hence, the delayed plateau phase of CTV⁺ cells in cDC1s was a reflection of the higher stability of the fluorochrome in cDC1s compared with macrophages, confirming earlier reports (47).

To investigate a potential link between the process of AC engulfment and homeostatic maturation, we compared the engulfment capacity of immature CCR7⁻ and mature CCR7⁺ cDC1s (Fig. 1D). This revealed that the proportion of CTV⁺ cells first increased in immature CCR7⁻ cDC1s temporally followed by an increase in CTV⁺ mature CCR7⁺ cDC1s. Because all the injected CTV⁺ ACs

would have been consumed by that time (fig. S4A), this suggested that CTV⁺CCR7⁺ cDC1s had derived from CTV⁺CCR7⁻ cells, in line with earlier data (26). Supporting this conclusion, injection of ACs led to an increase in the proportion of mature CCR7⁺ cDC1s (Fig. 1E), which was exclusively caused by an increase in CTV⁺CCR7⁺ cells (Fig. 1F, green line), because the numbers of CTV⁻CCR7⁺ cDC1s (Fig. 1F, red line) remained stable throughout the time course. The number of CCR7⁺ cDC2s did not change (Fig. 1E). The expression levels of CCR7 steadily increased in CTV⁺ CCR7⁺ cDC1s over time, showing that engulfment coincides with the onset of DC maturation (Fig. 1G). To establish a causal link between AC engulfment and homeostatic maturation, we analyzed *Rac1*^{fl/fl} × *Itgax*-Cre mice (termed *Rac1*ΔDC) that lack the small

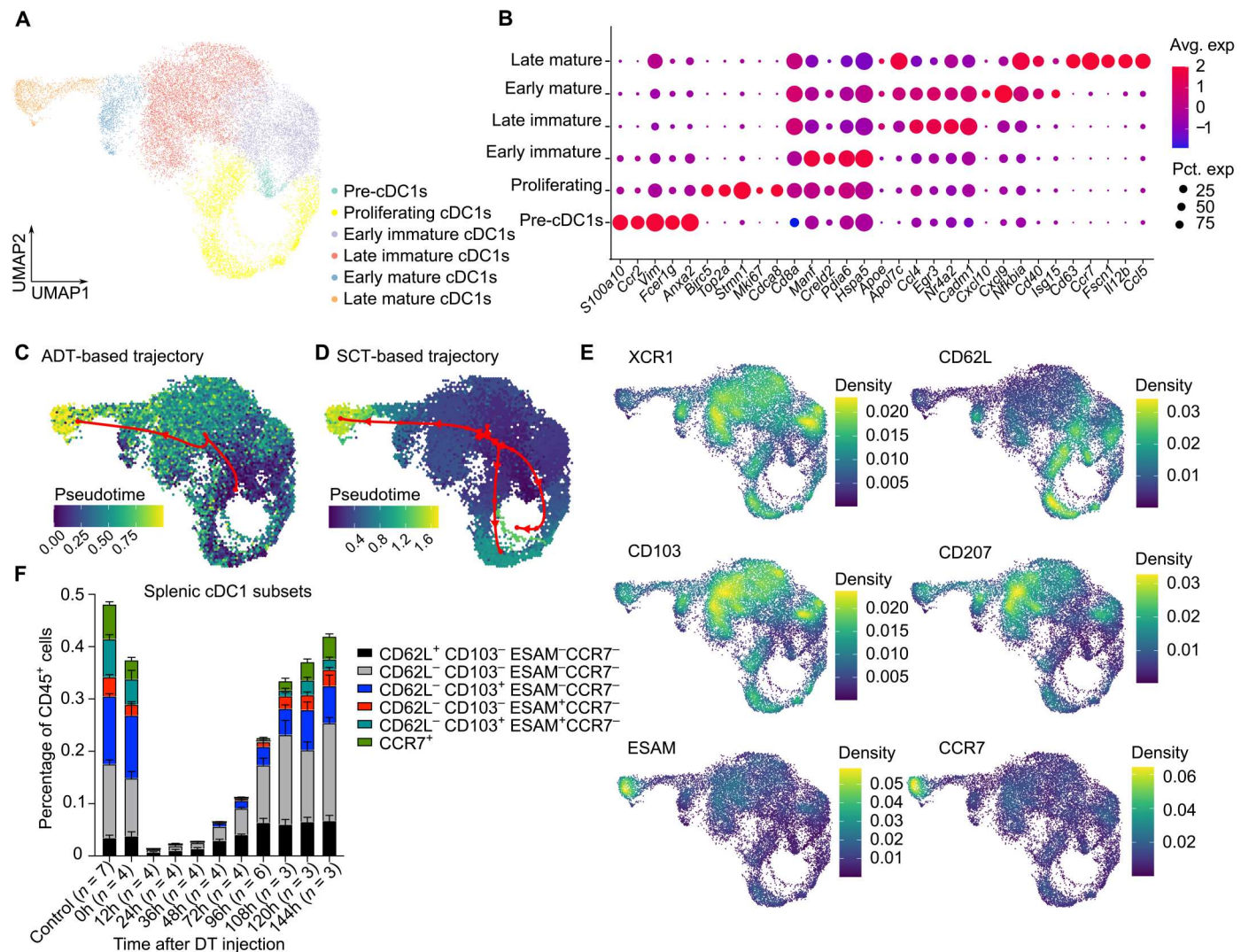


Fig. 2. CITE-seq allows for in-depth mapping of homeostatic maturation pathways in cDC1s. (A) UMAP plot of CITE-seq data of cDC1s showing annotated clusters. (*n* = 2) (B) Dot plot showing per cluster the expression of selected genes. The percentage of cells expressing the gene and the average expression of that gene within a cluster are represented by the dot size and color, respectively. (C and D) UMAP plots of CITE-seq data of cDC1s showing the trajectory inferred by SCORPIUS on ADT (protein) data in (C) and PAGA Tree on SCT (mRNA) data in (D). The red line shows the projected trajectory. The color scale represents the pseudotime variable. (E) UMAP Nebulosa plots showing the kernel density estimated expression of indicated ADT markers. The color scale represents the density estimates. (F) Stacked bar plot of cDC1 subsets from the spleens of CD45.1 mice, reconstituted with CD45.2 bone marrow of either *Xcr1*-Cre^{+/+} *Rosa26*-LSL-iDTR^{Tg/+} or *Xcr1*-Cre^{Tg/+} *Rosa26*-LSL-iDTR^{Tg/+} mice, at different time points after a DT (500 ng) intraperitoneal injection. The control mice (*Xcr1*-Cre^{+/+}) were injected with DT for 12 hours before euthanasia. The mean ± SEM is shown. Representative of two experiments in (A to F).

guanosine triphosphatase (GTPase) *Rac1* in the DC compartment and *Rac1* Δ DC *Rac2*^{-/-} mice that lack both *Rac1* and the partially redundant *Rac2* GTPase in DCs (fig. S4, B and C). RAC1 controls actin recruitment to engulfment receptors (48), and the engulfment of ACs by cDC1 was reduced in *Rac1* Δ DC mice (Fig. 1H). Consistent with the redundancy with RAC2, the effect on engulfment was more prominent in *Rac1* Δ DC *Rac2*^{-/-} mice (Fig. 1H). The block in AC engulfment led to a profound defect in cDC1 maturation, whereas cDC2 maturation remained unaffected (Fig. 1I and fig. S4D). Consistent with this, cDC1 maturation was not induced by injecting ACs in *Rac1* Δ DC/*Rac2*^{-/-} mice (Fig. 1J). Injection of *Rac1* Δ DC/*Rac2*^{-/-} mice with pIC resulted in the formation of a CCR7⁺ cDC1 population, showing that immunogenic cDC1 maturation was not affected by *Rac1* and *Rac2* deficiency (Fig. 1K). Thus, AC uptake promoted homeostatic cDC1 maturation, whereas homeostatic cDC2 maturation appeared to occur independently of AC uptake.

In-depth mapping of homeostatic maturation pathways in cDC1s using cellular indexing of transcriptomes and epitopes sequencing

To address how engulfment of ACs drives homeostatic maturation of cDC1s, we exploited the unique properties of the spleen where, in contrast to the periphery, DC maturation leads to intrasplenic migration from the red pulp and marginal zone to the white pulp (3, 29, 36, 49). Therefore, all DC maturation steps are present and may be studied within the same organ. We used cellular indexing of transcriptomes and epitopes by sequencing (CITE-seq) to map the homeostatic maturation process of cDC1s using a panel of 160 oligo-conjugated antibodies recognizing surface proteins (data file S2). CITE-seq was performed on CD64⁻ CD11c⁺ MHCII⁺ cDCs sorted from the spleens of nonimmunized mice in a spiked ratio of 70% cDC1s (XCR1⁺) and 30% cDC2s (CD172a⁺). Unsupervised clustering and Uniform Manifold Approximation and Projection (UMAP) dimensionality reduction yielded several clusters of cDC1s (fig. S5A) that were annotated on the basis of the expression of data-driven marker genes. From this, we identified pre-cDC1s, proliferating cDC1s, early immature cDC1s, late immature cDC1s, early mature cDC1s, and late mature cDC1s (Fig. 2, A and B, and data files S3 and S4). For consistency, we defined "immature" cDC1 subsets as CCR7⁻ and "mature" cDC1 subsets as CCR7⁺, and the terminology "early" or "late" refers to their relative position within the immature or mature subsets. By applying trajectory inference algorithms on antibody-derived tag (ADT) expression or on mRNA expression [normalized via Single Cell Transform (SCT)], a trajectory could be assigned that started at the pre-cDC1 stage and followed a number of differentiation steps up to late mature cDC1s (Fig. 2, C and D). Using plots generated with the Nebulosa package, we found that specific surface markers defined the transition steps between the different clusters (data file S5 and Fig. 2E). CD62L expression was associated with the more immature cDC1 subsets and was prominently expressed in pre-cDC1s, proliferating cDC1s, and early immature cDC1s (Fig. 2E). CD207 and CD103 were up-regulated during the transition from early immature to late immature cDC1s before cDC1s started to mature and express CCR7. The transition from late immature to early mature cDC1s was marked by the induction of a type I IFN signature, intermediate expression of XCR1, very low expression of CCR7 protein, and intermediate expression of several

maturation genes (Fig. 2, B and E; fig. S5C; and data files S3 to S5). CCR7 increased at the mRNA level in early mature cDC1s and became highly abundant at the protein level in late mature cDC1s, concomitant with a decline in XCR1 levels and expression of CD63 (also called lysosomal-associated membrane protein 3, LAMP3). Endothelial cell-specific adhesion molecule (ESAM), a marker typically used to subdivide cDC2s [fig. S6 (A and B) and (50, 51)], was especially prominent in mature cDC1s but was also found in late immature cells.

Next, we used this same set of markers (CD62L, ESAM, CD103, and CCR7) to immunophenotype cDC1s by flow cytometry and found the same heterogeneity among immature cDC1s (fig. S7, A and B). By designing a cDC1 panel that included these markers (gating strategy in fig. S2), we distinguished five distinct immature cDC1 subsets (CD62L⁺CD103⁻ESAM⁻, CD62L⁻CD103⁻ESAM⁻, CD62L⁻CD103⁺ESAM⁻, CD62L⁻CD103⁻ESAM⁺, and CD62L⁻CD103⁺ESAM⁺) and two mature subsets (CCR7^{int} and CCR7^{hi}) by manual gating (fig. S2 and S7B). Plotting the median fluorescence intensities (MFIs) for the maturation markers (MHCII, CD86, CD40, and CCR7) in each of the cDC1 subsets provided further evidence that they represented distinct steps along the homeostatic maturation gradient (fig. S7, C to E). To validate this, we used *Xcr1*-Cre x *Rosa26*-LSL-iDTR mice, which express the diphtheria toxin receptor (DTR) in all cDC1s, and immunophenotyped cDC1 clusters over time after DT-mediated deletion. In noninjected (0 h) or control-treated (*Xcr1*-Cre^{+/+}) mice, we detected all immature and mature cDC1 clusters using our cDC1 staining panel (Fig. 2F). In mice treated with DT, for up to 36 hours after injection, only immature CD62L⁺CD103⁻ESAM⁻ cDC1s and CD62L⁻CD103⁻ESAM⁻ cDC1s were detectable, marking them as early immature cDC1s. Gradually, we detected cDC1s expressing CD103, and 24 hours later, we detected populations of cDC1s expressing ESAM. At 108 hours after DT treatment, we could detect the late immature CD62L⁻CD103⁺ESAM⁺ subset and mature CCR7⁺ cDC1s. These data confirmed the ranking of the different cDC1 subsets determined by the surface expression of maturation markers (fig. S7E) and the trajectory inferred by bioinformatics algorithms (Fig. 2, C and D). Although we refer to the distinct clusters as "cDC1 subsets", these "subsets" refer to distinct maturation stages or states within the cDC1s rather than implying ontogenetically distinct subsets.

To identify the localization of different cDC1 subsets within the spleen, we performed an intravenous labeling approach where we used an anti-CD11c antibody to mark blood-exposed cells in the red pulp and marginal zone of the spleen by flow cytometry (49). All immature cDC1 subsets were highly labeled, indicating that they preferentially localized in the red pulp/marginal zone, with the late-immature cDC1s (CD62L⁻CD103⁺ESAM⁺) being almost exclusively localized there (fig. S7, F and G). Mature cells expressing CCR7 had limited labeling with intravenous anti-CD11c antibody, indicating that they were not localized within the blood-accessible red pulp and may have migrated to the white pulp (fig. S7, F and G). Thus, these data suggest that, as with immunogenic maturation, intrasplenic homeostatic maturation is associated with migration of cDC1s from the red to the white pulp, and during this process, cDC1s express discrete surface markers that can be used to follow their maturation.

Different cDC1 clusters are characterized by distinct gene signatures and function

To understand the biological processes driving the transitions between the distinct cDC1 maturation stages, we analyzed the gene expression signatures of each cluster and generated a heatmap representing the top DE genes in each cluster (Fig. 3A). **Pre-cDC1s** were characterized by genes such as *Ccr2*, *Fcer1g*, and *Cd24a* that are associated with a subset of primed pre-cDC1s (Fig. 3, A and B) (6, 52). In line with earlier observations (53–55), early committed cDC1s underwent cell cycle proliferation in situ ("**proliferating cDC1s**"), as revealed by the expression of cell cycle genes that allowed us to assign cells to the G₁, S, or G₂-M phase (Fig. 3, A and B, and fig. S5B). When exiting the cell cycle, several unfolded protein response (UPR) genes were induced, such as *Hspa5*, *Pdia3*, *Dnajc3*, or *Creld2* [Fig. 3, A and B, and fig. S5, A and C (cluster 0)]. "Activation of the UPR" was a characteristic gene signature of the cDC1 subset "**early immature cDC1s**," which was defined by the presence of *CD62L*.

During the transition from early immature cDC1s to **late immature cDC1s**, cDC1s became *CD207*⁺ *CD103*⁺ (data file S5), two markers associated with AC engulfment (29, 56, 57). At the gene expression level, the top gene that marked the transition from early to **late immature cDC1s** was *Apol7c*, an apolipoprotein with unknown function in DCs (data file S5 and Fig. 3, A and E) (58). In addition, *Apoe*, a well-established mediator of cholesterol efflux, was increased in expression, whereas *Fdps*, a rate-limiting enzyme for cholesterol biosynthesis, decreased in expression between early to late immature cDC1s (Fig. 3, A and E, and data file S5). Other genes that marked the transition from early to late immature cDC1s were a group of nuclear receptor proteins (*Nr4a2* and *Nr4a3*) and some early response genes (*Egr3* and *Egr2*; Fig. 3, A and B, and fig. S5C). Although the function of these genes in DCs is not fully established, *NR4A3* has been implicated in DC migration, and both EGR and NR4A proteins have been proposed to be negative regulators of DC immunogenicity (59–61). In macrophages, NR4A is induced upon engulfment as immunosuppressive factor (62). In addition, *Dnase1l3*, which encodes an enzyme required for degrading self-DNA from ACs and associated with autoimmunity (63), was up-regulated in late immature cDC1s compared with early immature cDC1s (Fig. 3A). We found that *Dnase1l3* was down-regulated in DCs from *Rac1ΔDC* *Rac2*^{-/-} mice, suggesting that *Dnase1l3* may be induced by apoptotic cargo (fig. S4D).

We confirmed a correspondence between cDC1 gene signatures and function, because we found that late immature *CD62L*⁻ *CD103*⁺ *ESAM*⁺ *CCR7*⁻ cDC1s showed the highest engulfment capacity (fig. S7H), which was consistent with their expression of the engulfment receptor *CD36* (fig. S7I). Once cDC1s matured and expressed *CCR7* on their surface, they down-regulated their phagocytotic capacity (64, 65), which was accompanied by a decrease in *CD36* expression. Increased engulfment capacity correlated positively with increased cellular granularity and cell size, as measured by flow cytometry side scatter (SSC) and forward scatter (FSC) intensity, respectively (fig. S7, J to L). The high engulfment capacity, cell granularity, and cell size of late immature cDC1s were observed both in the presence (fig. S7, H and L) or absence (fig. S7, I to K) of exogenously injected *CTV*⁺ ACs, consistent with the maturation of cDC1s taking place in response to the continuous engulfment of endogenous ACs.

The transition from late immature to early mature cDC1s was characterized by induction of IFN-inducible genes such as *Cxcl10*, *Cxcl9*, *Iigp1*, *Ifi47*, *Gbp2*, and *Gbp5* (Fig. 3A and fig. S5C), none of which were detected in cDC2s (fig. S6B). Several of these genes are not only associated with antiviral immunity but also become induced after AC engulfment in macrophages (66) or after uptake of extracellular DNA in cDC1s in a cyclic GMP-AMP synthase/stimulator of IFN genes (cGAS/STING)-dependent manner (67, 68). In support of this notion, *CXCL9* expression was increased in *CTV*⁺ cDC1s 2 hours after uptake of ACs, and STING deficiency led to a lower frequency of *CXCL9*⁺ cDC1s (fig. S8, A and B). The IFN signature was assigned to a defined subset of cells within the cluster of early mature cDC1s [cluster 16; fig. S5 (A and C)]. In cells where the *Ccr7* gene was expressed, the IFN signature was no longer detected (fig. S5C), suggesting that either expression of IFN genes was turned off or that this represented a distinct differentiation step of cDC1s not linked to final maturation. In mice lacking STING, the number of mature cDC1s was not affected (fig. S8C), suggesting that the STING pathway and IFN signature was not a prerequisite for homeostatic cDC1 maturation. Last, the transition to late mature cDC1s was characterized by the expression of several well-established markers such as *Cd63*, *Fcscn1*, *Il4i1*, or *Socs2*, in line with our earlier findings (fig. S1D and Fig. 3, A and B).

Thus, whereas early committed cDC1s underwent proliferation and a transient UPR response, the acquisition of AC engulfment receptors and concomitant increase in engulfment capacity marked a transition to *Nr4a* nuclear receptor expression and expression of cholesterol efflux and metabolism genes. Early in the process of maturation, cDC1s were marked by a transient type I IFN response, which was turned off in cells expressing *CCR7* and white pulp-migrating late mature cDC1s.

Homeostatic cDC1 maturation is characterized by cholesterol efflux

Earlier studies performed to define gene signatures distinguishing homeostatic from immunogenic mature cDC1s identified "super-pathway of cholesterol biosynthesis" as an IPA-derived signature specifically associated with homeostatic mature cDC1s (26). Using the DoRothEA regulon collection and VIPER algorithm (which infers transcription factor regulatory activity from single-cell transcriptomics data), we inferred high activity of *LXRβ*, a transcription factor that promotes the expression of genes required for cholesterol efflux from cells, from the late immature cDC1 stage onward (Fig. 3, C and D). Consistent with this, the well-established *LXR* pathway target genes *Abcg1* and *ApoE* showed increased expression as cDC1 cells matured (Fig. 3E). In line with earlier literature (26), we also identified that target genes of the sterol regulatory element-binding protein 2 (*SREBP2*) pathway, which promotes cholesterol uptake and biosynthesis within cells, such as *Ldlr*, *Fdps*, or *Hmgcr* became highly expressed in the mature cDC1 subsets (Fig. 3E).

Because our data indicated that genes associated with cholesterol homeostasis were induced as cDC1s progressed along the maturation gradient, we investigated cholesterol levels within DC populations. First, we sorted immature (*CCR7*⁻) and mature (*CCR7*⁺) cDC1s and cDC2s from the spleen and determined their total cholesterol levels (free and esterified) by an enzymatic approach. Already at baseline, cDC1s had higher cholesterol levels than cDC2s, and during maturation, their cholesterol levels further

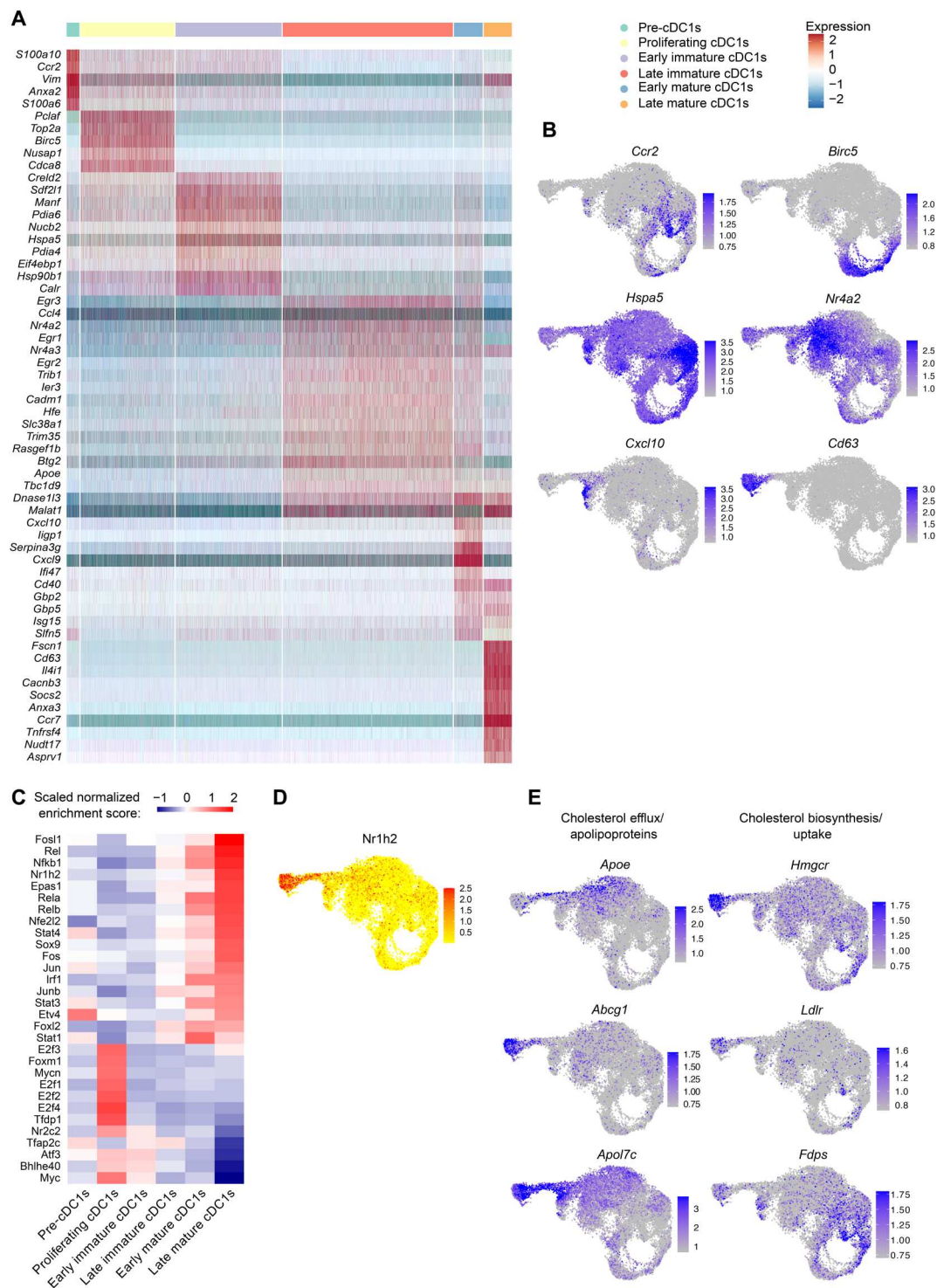


Fig. 3. Distinct gene signatures hallmark the different cDC1 clusters. (A) Heatmap showing the normalized expression of the top defining genes for each cDC1 cluster. The color bar at the top indicates the cDC1 cluster. The color scale represents the normalized gene expression. (B) UMAP feature plots of representative RNA markers for each cDC1 cluster. (C) Heatmap of the most variable transcription factors (TFs) across the cDC1 subclusters. The color scale of the heatmap represents the scaled TF activity (inferred from the mRNA expression of the TF targets). The columns indicate the cDC1 subclusters and are ordered according to maturation. The rows indicate the TFs and are ordered according to activity in the late mature cDC1s. (D) UMAP feature plot of the TF LXRβ (*Nr1h2*). The color scale represents inferred TF activity from LXRβ's transcriptional targets. (E) UMAP feature plots of genes involved in cholesterol metabolism. The color scale represents normalized gene expression. Representative of two experiments in (A) to (E).

increased, whereas the cholesterol levels in cDC2s remained relatively low (Fig. 4A). Moreover, cellular cholesterol levels were not up-regulated in immunogenic mature cDC1s elicited in response to pIC (Fig. 4B). We analyzed total cellular cholesterol in cDC1 subsets across maturation stages and found that cholesterol levels increased during the transition from early immature toward late immature cDC1s (Fig. 4C), which correlated with their increase in engulfment potential (fig. S7H). We confirmed these data using boron-dipyrromethene (BODIPY) and quantitative flow cytometry to measure neutral lipid levels, including esterified cholesterol, in each of the cDC1 subsets (Fig. 4D). In cDC1 cells from *Rac1ΔDC* *Rac2*^{-/-} mice, the BODIPY levels in the immature subsets were low (Fig. 4E), suggesting that the increased cholesterol may be linked to the ability to engulf dead cells. To investigate this, we injected mice with CTV-labeled ACs and used BODIPY to evaluate cholesterol levels in cells that had engulfed the dead cells. Consistently, we found that CTV⁺ cDC1s had higher BODIPY signals compared with CTV⁻ cDC1s 2 hours after injection with CTV-labeled ACs (Fig. 4F). Notably, cholesterol levels declined during maturation (Fig. 4, C to F) and returned to basal levels again 12 hours after AC injection (Fig. 4F). To determine whether the increase in cholesterol in CTV⁺ cDC1s may be derived from engulfed AC, we injected mice with ACs loaded with TopFluor-cholesterol. We found that CTV⁺ cDC1s showed high levels of fluorescence associated with TopFluor-cholesterol 2 hours after injection (Fig. 4G). In contrast, cDC2s did not engulf TopFluor-cholesterol-loaded ACs (Fig. 4G). Similar to what we noted before, 12 hours after injecting TopFluor-cholesterol-loaded ACs, the intensity of the TopFluor-cholesterol levels in CTV⁺ cDC1s had decreased.

In macrophages, it is established that engulfment of ACs increases the influx of cholesterol into cells, which dampens cholesterol biosynthesis and activates the LXR pathway, leading to increased expression of cholesterol efflux genes such as *Abca1* (69). Using quantitative polymerase chain reaction (qPCR), we determined that expression of genes such as *Abcg1*, the main cholesterol transporter in DCs (70), and *ApoE*, which is needed for reverse cholesterol transport, increased during the maturation of cDC1s (Fig. 4H), confirming our CITE-seq data (Fig. 3E). We also determined that the expression of the apolipoproteins *Apol7c* and *Apol10b* was higher in mature cDC1s (Fig. 4H), consistent with our CITE-seq analyses (data file S5 and Fig. 3, A and E) and previous work (58). Furthermore, whereas the genes encoding the cholesterol biosynthesis enzyme *Fdps* and *Sqle* and the low-density lipoprotein receptor *Ldlr* were relatively low in the late immature CD103⁺ESAM⁺ cDC1s, they were increased during the final step of maturation (Fig. 4H), as we had observed in our CITE-seq data (Fig. 3E). Thus, AC engulfment leads to a marked but transient increase in cholesterol levels due to the activation of cholesterol efflux genes.

Uptake of nonadjuvanted cholesterol-containing LNPs triggers homeostatic cDC1 maturation in vivo

Because the engulfment of ACs promoted homeostatic cDC1 maturation (Fig. 1E), and this correlated with accumulation of intracellular cholesterol, we investigated whether the process of engulfment itself or the uptake of lipids would drive the maturation of cDC1s. To address this, we injected mice with either inert fluorochrome-labeled latex beads, ACs, or cholesterol-containing LNPs and evaluated the maturation of cDC1s. Engulfment of latex beads did not induce cDC1 maturation in vivo, whereas engulfment of CTV-

labeled ACs did (Fig. 5A). LNPs consisted of 40% cholesterol, ionizable cationic lipids, phospholipids, and PEGylated lipids (71, 72). Traditionally, LNPs are coupled to TLR ligands such as pIC to induce immunogenic DC maturation (red dots in the right panel of Fig. 5A). However, uptake of empty LNPs (eLNPs) also triggered induction of cDC1 maturation, albeit with delayed kinetics compared with pIC-coupled LNPs (Fig. 5A). cDC2s, which did not show substantial engulfment of eLNPs, showed a small increase in maturation (fig. S9, A and B). Both the uptake of ACs (Fig. 5B) and the uptake of eLNPs (Fig. 5C) correlated with a down-regulation of CD80 expression in LNP⁺ CCR7⁺ cDC1s over time. Conversely, uptake of pIC-coupled LNPs increased CD80 expression (Fig. 5C and fig. S9C). Previously, cationic lipids present in LNPs have been proposed to activate TLRs, driving an inflammatory response and possibly an immunogenic maturation program in DCs (71, 73). However, neither eLNP-induced nor AC-induced cDC1 maturation was inhibited in *TRIF*^{-/-}*MyD88*^{-/-} mice, where all TLR-dependent signaling was blocked (Fig. 5D). Furthermore, qPCR analysis of LNP⁺ cDC1s isolated at 2 and 8 hours after injection of eLNP or pIC-LNPs confirmed that the tolerogenic DC maturation program was induced upon eLNP injection (Fig. 5E). Common markers of DC maturation, such as *Fscn1*, *Ccr7*, *Cd40*, *Cd80*, and *Cd86* (26), were up-regulated in cDC1s undergoing homeostatic-, eLNP-, or pIC-LNP-induced maturation (Fig. 5F), although at later time points, the levels of costimulatory molecules remained higher in cDC1s from mice injected with pIC-LNP. Genes previously associated with the homeostatic DC maturation program such as *Scube3*, *Slco5a1*, *Tmem176a*, or *Eno3* (26) were induced specifically in homeostatic cDC1s and in cDC1s that took up eLNPs but were not increased in cDC1s after uptake of pIC-LNPs. *Icosl* appeared specifically induced in homeostatic cDC1s and cDC1s that took up eLNPs. Only cDC1s that took up pIC-LNPs showed high expression of type I IFN-stimulated genes (Fig. 5F), typically associated with immunogenic mature DCs (26). On the basis of this gene expression analysis, we conclude that uptake of eLNPs in cDC1s induced a homeostatic DC maturation program, whereas uptake of pIC-LNPs induced an immunogenic DC maturation program.

Last, we assessed the expression of cholesterol metabolic genes in cDC1s after uptake of ACs, eLNPs, or pIC-LNPs because all conditions were expected to increase intracellular cholesterol levels. Cholesterol efflux genes and cholesterol metabolic genes were high in homeostatic mature cDC1s and in mature cDC1s induced by eLNPs but not in immunogenic mature cDC1s induced by pIC-LNPs (Fig. 5F). In line with this, uptake of eLNPs led to a transient increase in cholesterol levels in cDC1s, whereas uptake of pIC-coupled LNPs led to a sustained cholesterol increase (Fig. 5G). These data suggest that in immunogenic conditions, the LXR pathway and cholesterol efflux pathways are suppressed, consistent with observations of macrophages (74).

To test whether LNPs could induce cDC1 maturation in the periphery and the spleen, we injected LNPs in the quadriceps of mice and evaluated cDC1s from the draining iliac LN (dLN) 24 hours later. Both eLNP- and pIC-LNP-loaded mature cDC1s were found in the dLN but were not detected in a non-draining LN (fig. S9, D to F). Similar to what we observed in the spleen, cDC1s that engulfed pIC-LNPs had higher cholesterol levels (fig. S9G) and showed increased CD80 and CD86 expression (fig. S9H) compared with cDC1s that took up eLNPs.

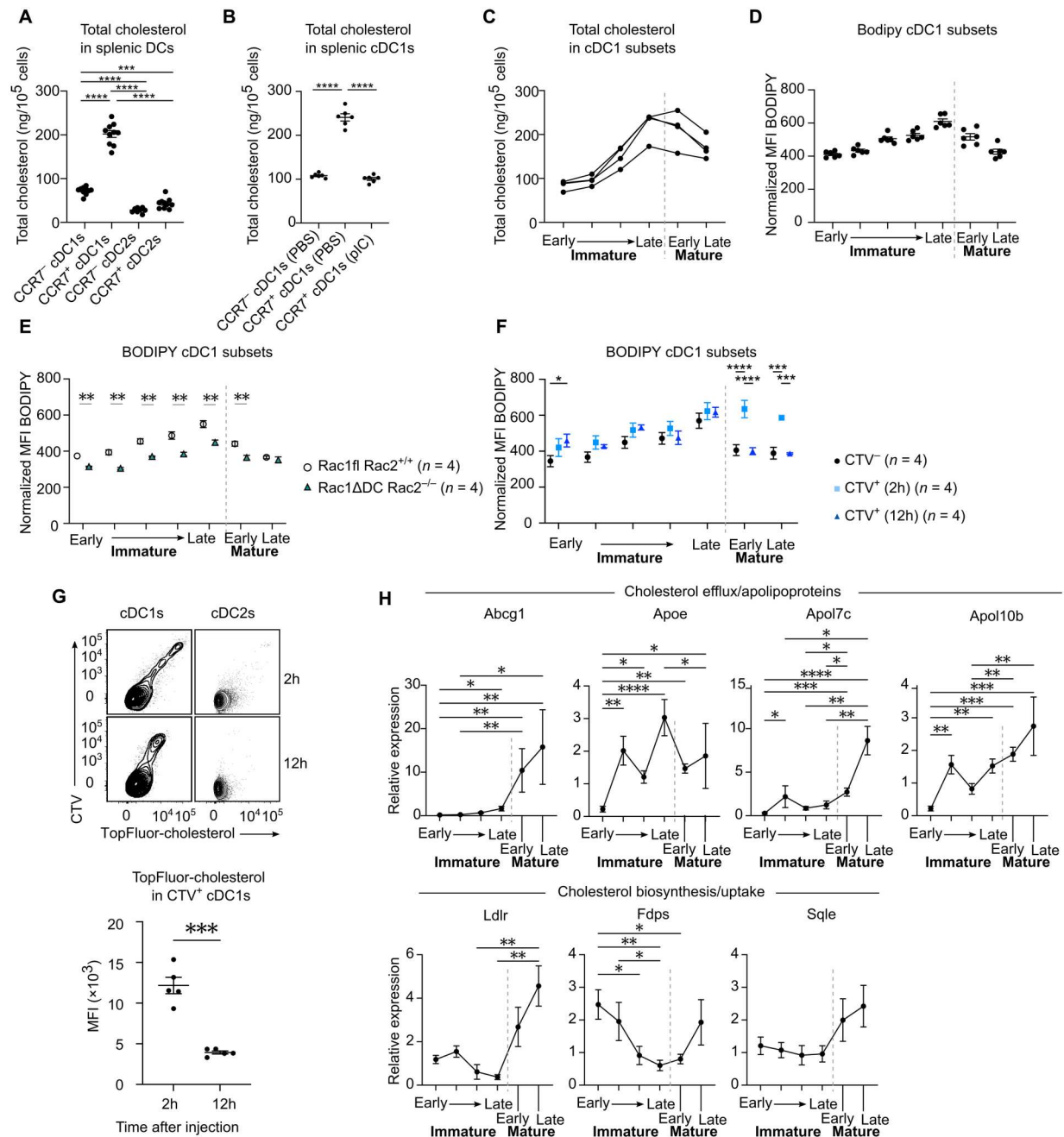


Fig. 4. Homeostatic cDC1 maturation is characterized by cholesterol efflux. (A) Total cholesterol levels determined by enzymatic assay from sorted CCR7⁺ and CCR7⁻ cDC1s and cDC2s at steady state. One-way ANOVA, corrected for multiple testing by Tukey's multiple comparisons test with a single pooled variance. (B) Total cholesterol levels determined by enzymatic assay from sorted CCR7⁺ and CCR7⁻ cDC1s after pIC injection (intraperitoneally, 20 μ g, 16 hours). One-way ANOVA, corrected for multiple testing by Tukey's multiple comparisons test with a single pooled variance. (C) Total cholesterol levels determined by enzymatic assay from sorted cDC1 subsets at steady state. Each line represents one individual mouse. (D) The normalized MFI of BODIPY 493/503 in cDC1 subsets. (E) The normalized MFI of BODIPY 493/503 in cDC1 subsets from the spleen of *Rac1^{fl/fl}Rac2^{+/+}* and *Rac1 Δ DC Rac2^{-/-}* mice. Two-way ANOVA, corrected for multiple testing by controlling the false discovery rate. Asterisks represent the *q* value. (F) The normalized MFI of BODIPY 493/503 in CTV⁻ and CTV⁺ cDC1 subsets 2 and 12 hours after intravenous injection of CTV-labeled ACs. Two-way ANOVA, corrected for multiple testing by Tukey's multiple comparisons test with a single pooled variance. (G) Engulfment of CTV-labeled TopFluor-cholesterol-loaded ACs in splenic cDC1s and cDC2s. Top: Representative contour plots showing CTV and TopFluor-cholesterol signal. Bottom: Bar graph showing the average TopFluor-cholesterol signal in CTV⁺ cDC1s 2 and 12 hours after uptake of ACs. Unpaired *t* test with Welch's correction. (H) Relative expression of cholesterol efflux, synthesis, and influx genes in indicated cDC1 subsets, measured by reverse transcription qPCR (RT-qPCR). The expression was normalized to housekeeping genes *Actb*, *Rpl13a*, *Sdha*, and *Gapdh*. Friedman test corrected for multiple testing by controlling the false discovery rate. Asterisks represent the *q* value. One outlier was removed for *Abcg1* and *Ldlr* after ROUT method of detecting outliers with *Q* = 0.1% (*n* = 8; *Abcg1*: *n* = 7; *Ldlr*: *n* = 7). The mean $\pm$ SEM is shown in (A and B) and (D to H). Representative of two experiments in (A) and (C to H). The *x* axes in (C to F) and (H) represent cDC1 subsets based on markers CD62L, CD103, ESAM, and CCR7 as defined in fig. S7. In (C and H), the CD62L⁻ CD103⁺ ESAM⁺ CCR7⁻ subset was not included. **P* < 0.05, ***P* < 0.01, ****P* < 0.001, and *****P* < 0.0001.

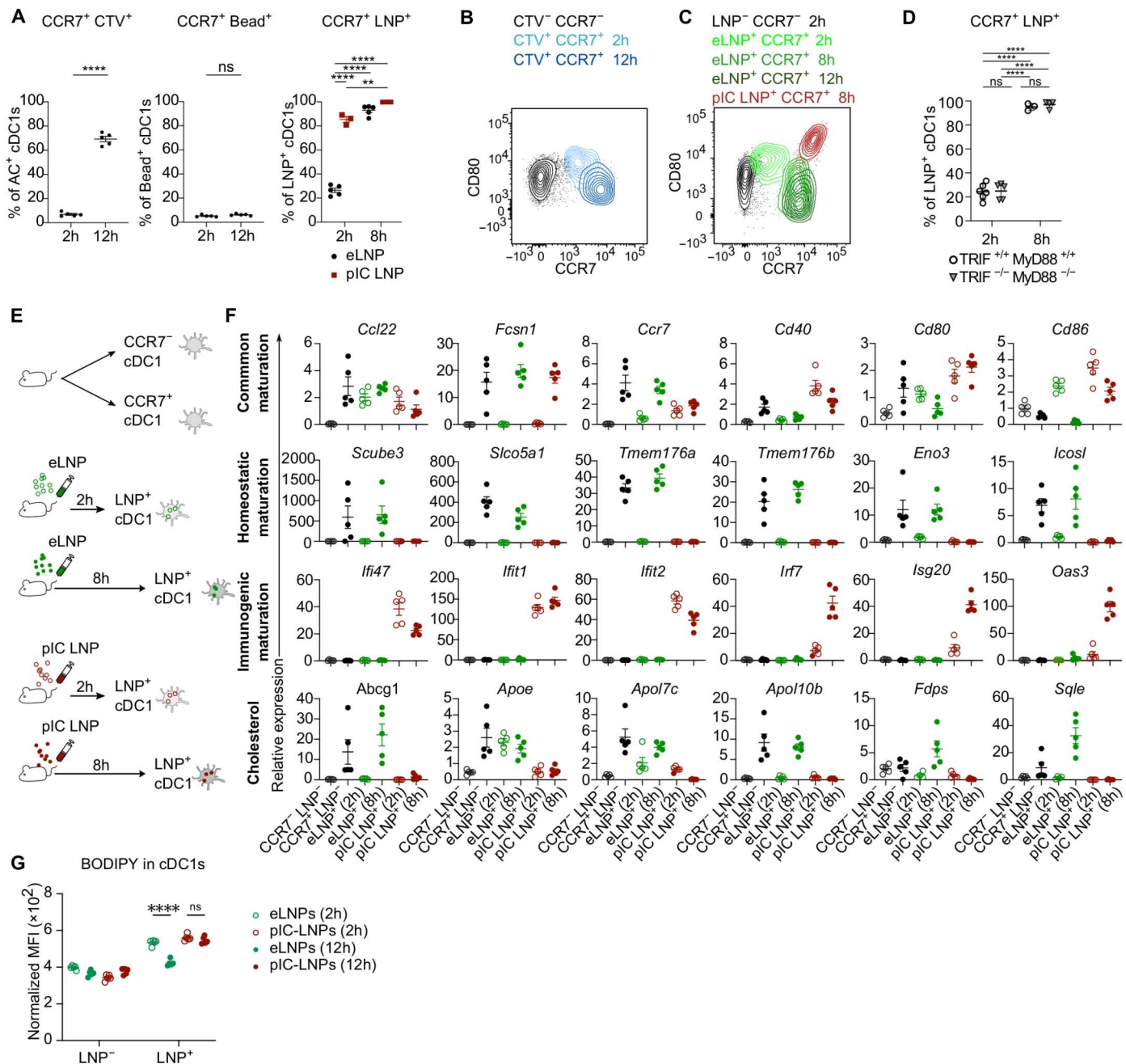


Fig. 5. Uptake of nonadjuvanted cholesterol-containing LNPs triggers homeostatic cDC1 maturation. (A) The frequency of CCR7⁺ cDC1s within AC⁺ cDC1s (left), bead⁺ cDC1s (middle), or LNP⁺ cDC1s (right) 2 and 12 hours after injection of CTV-labeled ACs and fluorescent beads or 2 and 8 hours after injection of Cy5-labeled empty or pIC-coupled LNPs. Unpaired *t* test with Welch's correction (ACs and beads). Two-way ANOVA, corrected for multiple testing by Šidák's multiple comparisons test with single pooled variance (LNPs). (B and C) Representative contour plots of flow cytometry data showing the expression of CD80 and CCR7 on indicated cDC1 subsets after CTV-labeled AC injection in (B) and Cy5-labeled LNPs in (C). (D) The frequency of CCR7⁺ cDC1s within LNP⁺ cDC1s in WT and TRIF^{-/-} MyD88^{-/-} mice 2 and 8 hours after injection of Cy5-labeled eLNPs. Two-way ANOVA, corrected for multiple testing by Tukey's multiple comparisons test with a single pooled variance. (E and F) Relative expression of common maturation markers (first row), homeostatic maturation markers (second row), immunogenic maturation markers (third row), and cholesterol efflux/biosynthesis genes (last row) in indicated cDC1 subsets of mice at steady state or after being intravenously injected with Cy5-labeled empty or pIC-coupled LNP for 2 or 8 hours, measured by RT-qPCR. The expression was normalized to housekeeping genes *Gapdh*, *Rpl13a*, and *Ywhaz*. (G) The normalized MFI of BODIPY 493/503 of LNP⁻ and LNP⁺ cDC1s from mice intravenously injected with Cy5-labeled empty or pIC-coupled LNP for 2 or 12 hours. Two-way ANOVA, corrected for multiple testing by Tukey's multiple comparisons test with a single pooled variance. Statistics are only shown for the LNP⁺ cDC1s. The mean ± SEM is shown in (A to G). Representative of two experiments in (A to G). ***P* < 0.01, ****P* < 0.001, and *****P* < 0.0001.

As we observed in AC-induced cDC1 maturation, uptake of eLNPs correlated with increased expression of SREBP2 target genes in late mature cDC1s (Fig. 5F). Simultaneously increasing the expression of cholesterol efflux and cholesterol synthesis genes seems counterintuitive but could be explained by the previously established role for ABCG1 shuttling oxysterols out of the endoplasmic reticulum, resulting in derepression of SREBP2 (75, 76). To address this, we generated *Abca1/Abcg1*ΔDC mice in which *Abcg1* is no longer induced upon maturation of cDC1s (fig. S10A). CCR7⁺ cDC1s from *Abca1/Abcg1*ΔDC mice no longer expressed the SREBP2 target genes *Fdps*, *Sqle*, and *Ldlr* (fig. S10B). Combined deficiency of *Abca1* and *Abcg1* led to the accumulation of neutral lipids and cholesterol esters in all cDC1 maturation stages, as monitored by BODIPY staining (fig. S10C). However, we did not detect any defect in homeostatic cDC1 maturation (fig. S10D). In mice where SREBP2 was conditionally knocked-out of DCs (*Srebp2*ΔDC mice), CCR7⁺ cDC1s did not express *Fdps*, *Sqle*, and *Ldlr* but did not exhibit defects in cholesterol levels or development of a maturation phenotype, except for a reduction in *Ccr7* expression levels (fig. S10, E to I).

In summary, tolerogenic cDC1 maturation was associated with the induction of a cholesterol efflux program initiated in late immature cDC1s. Increased expression of cholesterol efflux genes was accompanied by the activation of a SREBP2-dependent gene expression program during the final maturation step. In immunogenic conditions with concomitant TLR ligation, the LXR-dependent (and SREBP2-dependent) program was overridden, leading to sustained high cellular cholesterol levels.

Loss of LXRβ signaling leads to induction of an immunogenic maturation program in cDC1s

Increased intracellular levels of cholesterol in immune cells have been reported to increase risks for inflammatory diseases such as atherosclerosis or autoimmunity (77). In addition, it has been postulated that a decrease in cholesterol efflux in cDC1s would activate the inflammasome (70) and/or lead to increased T cell activation (78). To address whether cholesterol homeostasis contributes to the decision process between homeostatic and immunogenic cDC1 maturation, we generated mice where LXRα and LXRβ, the two main transcription factors controlling genes involved in the cholesterol efflux pathway (79), were deleted in DCs (fig. S11A). In wild-type (WT) cDC1s, only LXRβ (encoded by *Nr1h2*) was expressed, as revealed by CITE-seq analysis (fig. S11B). Deletion of both LXRα and LXRβ led to increased cholesterol levels in late immature and mature cDC1s (Fig. 6A) and decreased the survival of both the immature and mature subsets (fig. S11C). Analysis of costimulatory molecule expression in different cDC1 subsets revealed an increase in CD40 and CD86 expression levels, whereas CD80 expression levels appeared unaffected in response to deficiency of LXRα/β (Fig. 6A). Moreover, the surface expression of CCR7 appeared enhanced in LXRα/β-deficient DCs (fig. S11, D and E). These markers were not changed in expression at the mRNA level; on the contrary, *Ccr7* mRNA levels decreased in LXRα/β knockout cDC1s (Fig. 6B), suggesting that the stabilization of these proteins at the surface occurred at a posttranslational level. We observed a similar effect on *Ccr7* mRNA versus CCR7 protein levels when comparing immunogenic pIC-LNP-matured WT cDC1s with homeostatic eLNP-matured WT cDC1s (Fig. 5, C and F). Targeted analysis of gene expression in LXRα/β-deficient cDC1s

revealed a similar expression of common maturation genes, whereas the induction of most homeostatic genes appeared at least partially affected. Type I IFN response genes were increased by LXRα/β deficiency in both the mature and the immature subsets (Fig. 6B). Whereas the LXR target gene *Apoe* was no longer expressed in mature LXRα/β-null cDC1s, the expression of *Abcg1* appeared only mildly decreased (Fig. 6B). In line with this, the induction of the SREBP2 target genes *Fdps* and *Sqle* in homeostatic mature cDC1s was not affected by the absence of LXRα/β (Fig. 6B).

To investigate whether these observations were due to an overall decrease in the number of DCs in the LXRΔDC mice, we analyzed mixed bone marrow chimeras of CD45.1.2 WT bone marrow and CD45.2 LXRΔDC bone marrow (fig. S11, F and G). BODIPY staining confirmed an increase in neutral lipid and cholesterol ester content in late immature and mature cDC1s derived from LXRΔDC bone marrow compared with WT bone marrow (fig. S11H). Surface expression of the costimulatory molecules CD86 and CD40 was increased in cDC1 subsets derived from LXRΔDC bone marrow compared with WT bone marrow (fig. S11H). qPCR analyses found that homeostatic genes were decreased, and immunogenic genes were increased in CCR7⁺ cDC1s derived from LXRΔDC bone marrow compared with CCR7⁺ cDC1s from WT bone marrow (fig. S11I). Moreover, we found that *Apoe* expression was lost and *Abcg1* expression was not changed in cDC1s derived from LXRΔDC bone marrow compared with WT bone marrow (Fig. 6B), recapitulating what we observed with LXRΔDC mice. Thus, we established LXR signaling as an important switch to induce homeostatic DC maturation.

To test whether loss of LXR in DCs could break peripheral tolerance, we used an established model of tolerance induction to ovalbumin (OVA) (67, 80). We intravenously injected control (LXR^{fl/fl}) or LXRΔDC mice with 0.5 million naive OVA-specific CD8⁺ T cells (OT-I). One day later, the acceptor mice were injected either (i) with apoptotic thymocytes derived from gender-matched ActmOVA mice, (ii) with apoptotic thymocytes derived from gender-matched WT mice as a negative control, or (iii) with pIC-transfected apoptotic thymocytes from gender-matched ActmOVA mice as a positive control. Six days later, the acceptor mice were injected with a live cell mix of CTR-labeled ActmOVA splenocytes or CTV-labeled WT splenocytes in a 1:1 ratio. One day later, OT-I cytotoxicity was measured by determining the ratio of CTV-labeled (WT) versus CTR-labeled (OVA) cells (Fig. 6C), with a ratio of one indicating equal survival of both cell types. A ratio greater than one indicated preferential survival of CTV-labeled WT cells compared with CTR-labeled OVA-expressing thymocytes. LXR^{fl/fl} or LXRΔDC mice that received WT-ACs without OVA did not kill CTR- or CTV-labeled cells (Fig. 6D). Live OVA cells were relatively well tolerated in LXR^{fl/fl} mice that received OVA-ACs compared with mice that received pIC-transfected OVA-ACs (Fig. 6D). However, LXRΔDC mice that received OVA-ACs showed impaired T cell tolerance compared with LXR^{fl/fl} mice but did not show increased killing of pIC-transfected OVA-ACs (Fig. 6D). The overall survival of WT target cells was not affected by any treatment or by deletion of LXRα/LXRβ in DCs (fig. S11J). OT-I T cells from LXRΔDC mice that received OVA-ACs showed a significant increase in the surface expression of Ly6C compared with LXR^{fl/fl} OT-I T cells (Fig. 6E). Ly6C is a marker expressed by cytotoxic T cell lymphocytes (CTLs) (68, 81), which was also

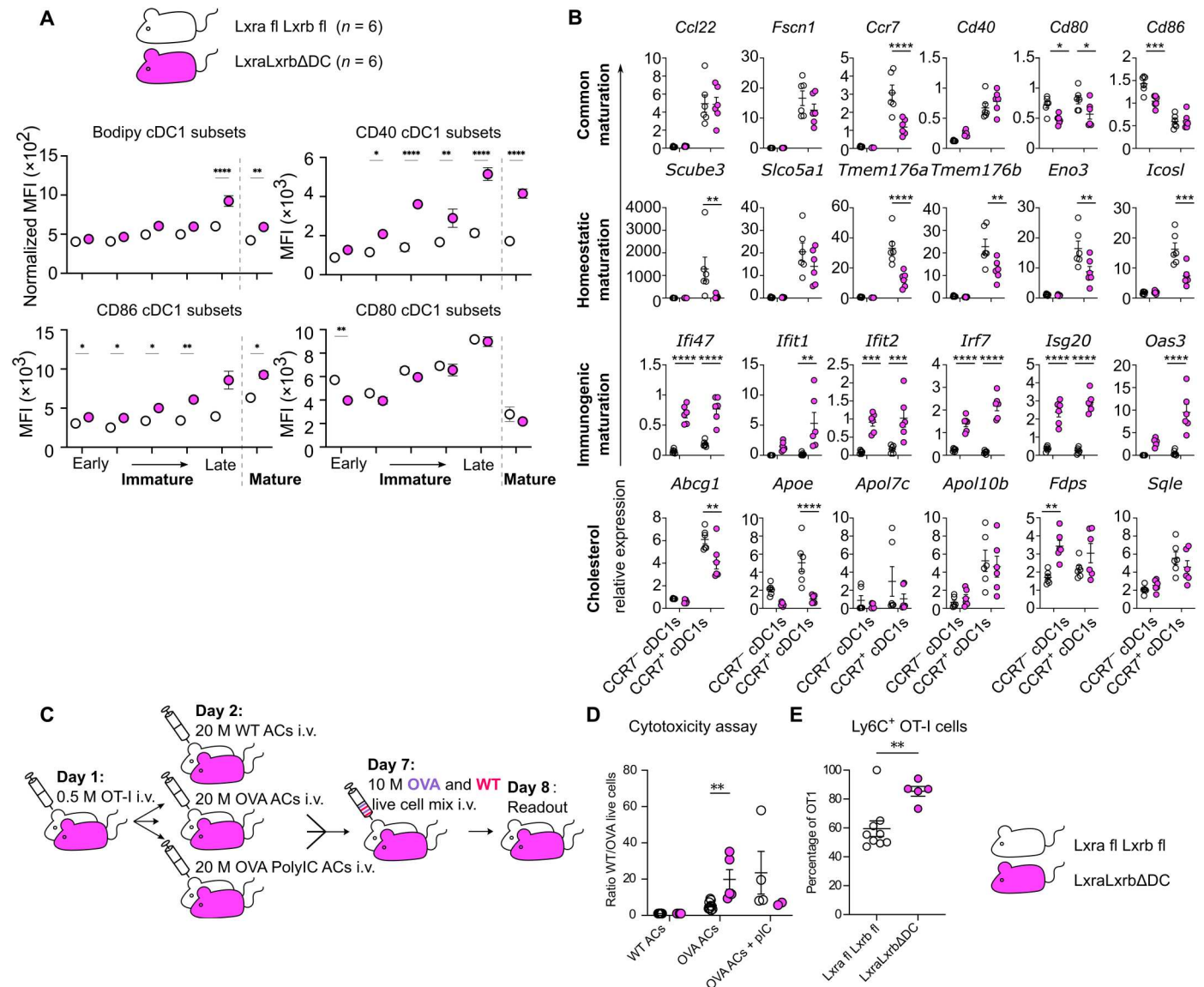


Fig. 6. Loss of LXR β signaling leads to induction of an immunogenic maturation program in cDC1s. (A) The normalized MFI of BODIPY 493/503 and MFI of costimulatory proteins (CD40, CD80, and CD86) on distinct cDC1 subsets in the spleen of LXR $\alpha^{fl/fl}$ LXR $\beta^{fl/fl}$ or LXR α LXR β ΔDC mice. Two-way ANOVA, corrected for multiple testing by Šidák's multiple comparisons test with a single pooled variance (BODIPY and CD40), individual variances (CD80 and CD86) (n = 6). (B) Relative expression of common maturation markers (first row), homeostatic maturation markers (second row), immunogenic maturation markers (third row), and cholesterol efflux/biosynthesis genes (last row) in CCR7⁺ and CCR7⁺ cDC1s of LXR $\alpha^{fl/fl}$ LXR $\beta^{fl/fl}$ /LXR α LXR β ΔDC mice. The expression was normalized to housekeeping genes *Gapdh*, *Rpl13a*, and *Ywhaz*. Two-way ANOVA, corrected for multiple testing by Šidák's multiple comparisons test with a single pooled variance. (C) Experimental model. CD45.2 acceptor mice (LXR $\alpha^{fl/fl}$ LXR $\beta^{fl/fl}$ or LXR α LXR β ΔDC) were injected with 0.5 million naïve CD45.1.2 OT-I cells. One day later, the acceptor mice were injected with either apoptotic thymocytes from ActmOVA mice, apoptotic thymocytes from WT mice as a negative control, or pIC-transfected apoptotic thymocytes from ActmOVA mice as a positive control. On day 7, the acceptor mice were injected with a 1:1 ratio mix of live cells from ActmOVA (CTR-labeled) and WT (CTV-labeled) mice. One day later, OT-I cytotoxicity was measured by determining the ratio of CTV-labeled (WT) versus CTR-labeled (OVA) cells. (D) OT-I cytotoxicity measured by the ratio of CTV- and CTR-labeled cells in LXR $\alpha^{fl/fl}$ LXR $\beta^{fl/fl}$ and LXR α LXR β ΔDC mice. Multiple Mann-Whitney U tests, corrected for multiple testing by controlling the false discovery rate. Asterisks represent the *q* value. (E) The percentage of Ly6C⁺ OT-I cells at day 8 after injecting OVA ACs at day 2. Unpaired *t* test with Welch's correction. The mean $\pm$ SEM is shown in (A to F). Representative of two experiments in (A and B). The schematic x axes in (A) represent cDC1 subsets based on markers CD62L, CD103, ESAM, and CCR7 as defined in fig. S7. **P* < 0.05, ***P* < 0.01, ****P* < 0.001, and *****P* < 0.0001.

elevated in expression on endogenous CD8 T cells in steady-state LXR Δ DC mice (fig. S12).

Extensive immunophenotyping of the lymphocyte compartment of 10-week-old LXR Δ DC mice revealed an increased percentage of activated B cells (CD86⁺ B cells) and an increased percentage of natural killer cells compared with that in LXR^{fl/fl} mice (fig. S12, A and B). CD4 and CD8 T cells were less naïve and expressed surface markers consistent with a central memory phenotype (fig. S12, C and D). The percentage of CD4 and CD8 T cells that expressed Ly6C was increased in LXR Δ DC mice compared with LXR^{fl/fl} mice (fig. S12, C and D). In comparison with LXR^{fl/fl} mice, LXR Δ DC mice had more T_{regs}. These T_{regs} also appeared more activated than those in LXR^{fl/fl} mice because there was a higher percentage of CD69⁺ and PD1⁺ T_{regs}, indicative of a switch to an effector memory phenotype (fig. S12E). Overall, this indicates an inflammatory phenotype in steady-state LXR Δ DC mice, in line with earlier data in literature (78). In summary, our data indicate that homeostatic maturation of cDC1s can be triggered in vivo by engulfment of ACs or nonadjuvanted LNPs and that distinct regulation of cholesterol metabolic pathways controls the maturation of homeostatic and immunogenic DCs (fig. S13).

DISCUSSION

In contrast to the detailed knowledge of PRR-induced immunogenic DC maturation under conditions of infection and inflammation, the molecular basis for homeostatic DC maturation has long been an enigma. Early studies excluded a major role for commensal microbiota or endogenous cytokine signaling as triggers for the homeostatic maturation process. Therefore, it has been proposed that "spontaneous" maturation occurs as a result of a random, stochastic process (26). At least for cDC1s, this appears an unlikely scenario because our data have revealed that the engulfment of ACs is necessary and sufficient to trigger homeostatic cDC1 maturation. The process of cDC1 maturation is accompanied by an increase in intracellular cholesterol levels, granularity, and cell size. In the spleen, once cDC1s start migrating to the white pulp, they lose their capacity to engulf ACs, and intracellular cholesterol levels decline. CCR7 expression steadily increases, peaking upon reaching the white pulp. We used CITE-seq analysis and trajectory algorithms to map the pathway of homeostatic cDC1 maturation and to understand the changes in transcriptome and surface expression markers upon transition from one maturation state to the next. Several of the signatures that were identified had been previously linked with metabolic adaptations to the influx of lipids, nucleotides, and other metabolites during the process of efferocytosis in macrophages. Homeostatic maturation of cDC1s could be mimicked by feeding DCs empty, nonadjuvanted LNPs, suggesting that uptake of exogenous lipids, rather than the process of engulfment, is sufficient to drive the tolerogenic maturation program.

In macrophages, one of the most established pathways that become activated upon efferocytosis is the LXR pathway, traditionally triggered not only by oxysterols but also by desmosterol, both of which accumulate in cells as a result of cholesterol influx (82, 83). Activation of the LXR branch activates the transcription of cholesterol efflux genes such as *Abca1*, *Abcg1*, and *Apoe* and leads to suppression of inflammation by activating TGF β 1 and interleukin-10 and/or dampening nuclear factor κ B (NF- κ B) and activating protein 1 proinflammatory target genes (84, 85). Loss of LXR α/β

leads to defects in AC engulfment, both in macrophages and in dendritic cells (84). On the basis of the induction of the cholesterol efflux genes *Apoe* and *Abcg1* in maturing cDC1s, we concluded that the LXR pathway was activated upon engulfment in cDC1s. Splenic cDC1s specifically express LXR β , whereas LXR α is hardly detected. In line with our hypothesis, loss of LXR α/β in cDC1s (LXR α/β Δ DC) led to higher cholesterol levels, especially in the late immature cDC1 subset. This is the subset where *Apoe* expression is maximal in WT cDC1s, which was completely abrogated in cDC1s from LXR α/β Δ DC mice. Notably, the cholesterol efflux during the transition from late immature to late mature cDC1s was not completely curtailed in LXR α/β double-knockout cDC1s, suggesting the existence of additional non-LXR-dependent efflux pathways. In this regard, we noticed that the cholesterol transporter *Abcg1* was still induced during cDC1 maturation in LXR α/β Δ DCs, in contrast to what we expected on the basis of literature data (79). During the very last phase of DC maturation, we found increased expression of SREBP2 target genes such as the cholesterol biosynthesis enzymes *Sqle* and *Fdps* and the lipoprotein receptor *Ldlr*, an observation that had been made before (26). Although it seems counterintuitive that both cholesterol efflux and synthesis are activated concurrently, these data are in line with the ability of ABCG1 to facilitate the activation of SREBP2 by shuttling of oxysterols out of the ER (75, 76). Analysis of cDC1s from *Abca1/Abcg1* Δ DCs mice was consistent with ABCG1 activating the SREBP2 pathway in late mature cDC1s.

In contrast to macrophages, the link between LXR activation and DC function remains poorly understood. Some studies coupled activation of LXR to a block in DC maturation and migration (86, 87), whereas others reported the opposite (88, 89). Almost all these studies were performed in in vitro generated monocyte-derived DCs that express LXR α rather than LXR β (86), in line with the notion that granulocyte-macrophage colony-stimulating factor cultures comprise a heterogeneous population of monocyte-derived macrophages and cDC2-like cells but not cDC1s (90). In splenic cDC1s, we found that loss of LXR α/β caused a reduction in *Ccr7* mRNA levels, whereas at the protein level, CCR7 levels at the cell surface increased, possibly caused by a posttranslational stabilization of the chemokine receptor due to increased levels of cholesterol in the plasma membrane, although this remains to be definitively tested. Similar observations were made for expression of the costimulatory molecules CD40 and CD86, whereas CD80 levels appeared unaffected. Although we detected mild effects on expression of homeostatic markers during maturation in LXR α/β -deficient cDC1s, expression of genes specifically linked to the immunogenic maturation program, such as *Oas3*, *Ifi47*, or *Irf7*, were activated, albeit not to the extent of what we observed during immunogenic, TLR-triggered maturation of cDC1s. This suggests that, as in macrophages (91), LXR pathways in cDC1s might be involved in basal repression of proinflammatory genes, in this way preventing immunogenic maturation. The relevance of the SREBP2 pathway in homeostatic mature DCs is less clear. We postulated that SREBP2 might be needed for the induction of the tolerogenic maturation program in cDC1s. However, analysis of *Srebp2*^{-/-} cDC1s did not support this hypothesis.

What is the molecular signal driving cDC1 maturation in homeostatic conditions? The engulfment of apoptotic material leads to an influx of a variety of nucleotides, lipids, proteins, sugars, and other metabolites, suggesting many possible triggers. We

noticed a potent but transient increase in type I IFN signature in cDC1s, but not in cDC2s, in a very defined subset of cells that were mostly CD103⁺CD207⁺ESAM⁺, corresponding to the subset with maximal engulfment potential. In tumors, recognition of tumor cell-derived DNA by engulfing DCs leads to the activation of the cytosolic cGAS/STING pathway and production of type I IFN (92), and we speculate that, similar to tumor DCs, recognition of self-DNA could trigger the activation of the type I IFN signature in early mature cDC1s. However, we did not detect major differences in DC maturation in STING-deficient cDC1s. Alternatively, lipids such as cholesterol or cholesterol-derived metabolites could incite cDC1 maturation. Along these lines, injection of LNPs comprising 40% of cholesterol induced cDC1 maturation in a TLR-independent manner. The LNP-induced activation of cholesterol metabolic pathways as the activation of tolerogenic genes was indistinguishable from the program induced by regular efferocytosis. Notably, coupling of the LNPs to pIC completely blocked cholesterol metabolic reprogramming, in line with earlier data on the role of TLR-induced interferon regulatory factor 3 in blocking the LXR response (74, 77). Although these data demonstrate that uptake of exogenous lipids in cDC1s is sufficient to incite the homeostatic maturation program, they do not reveal whether cholesterol or any other lipid species embedded within the LNPs (comprising 40% cholesterol, 50% ionizable cationic lipid, 10% phospholipids, and 1.5% PEGylated lipids) is the actual driver of maturation. A recent report established that phagosomal rupture could facilitate antigen cross-presentation in cDC1s (93), suggesting that the ionizable lipids present in LNPs, which ensure endosomal escape of the encapsulated cargo, could be a trigger of maturation.

Another major question is what defines the tolerogenic aspects of homeostatic mature cDC1s? Comparing gene signatures induced in cDC1s after uptake of eLNPs versus pIC-coupled LNPs confirmed that they induce distinct maturation programs (13, 26). Whereas uptake of LNPs or ACs led to rewiring of cholesterol metabolic programs, this was not the case after uptake of pIC-coupled LNPs, which caused a sustained rather than a transient increase in intracellular cholesterol levels. It has been postulated that cholesterol efflux in DCs is linked with immune tolerance. Concomitant loss of *Abca1* and *Abcg1*, the two main cholesterol efflux transporters, in DCs leads to an accumulation of cholesterol, activation of the inflammasome, and the development of autoimmune features (70). Similarly, loss of LXR α and β in conditions of hypercholesterolemia promotes cholesterol overload in antigen-presenting cells, leading to enhanced B cell-activating factor from the tumor necrosis factor family (BAFF) and a proliferation-inducing ligand (APRIL) production, enhanced antigen presentation, and development of autoimmune-related pathology (78). The data presented here suggest that LXR activation might be needed to turn off an immunogenic program in cDC1s and support the induction of a tolerogenic program, which could explain the enhanced antigen presentation and autoimmune phenotype in the absence of LXRs. In the presence of TLR ligands, the LXR pathway is suppressed, leading to a complete block in cholesterol efflux and an increase in inflammatory signaling, which could be relevant upon engulfment of virally infected cells. However, we found that the absence of the cholesterol efflux transporters *Abca1/Abcg1* did not lead to a similar induction of immunogenic genes, suggesting that the block in cholesterol efflux alone is not sufficient to explain the switch in maturation program in the absence of LXR β .

Last, although our dataset established a role for AC engulfment as a driver for cDC1 maturation, none of the phenotypic changes that we observed during cDC1 maturation were detected in cDC2s. It has been demonstrated that in a model of immunization with allogeneic red blood cells, which are exclusively engulfed by cDC2s (94), positioning of cDC2s in the spleen is strictly dependent on expression of Epstein-Barr virus-induced G protein-coupled receptor 2 (EBI2) and the presence of oxysterol gradients (95). Hence, it appears that both cDC1s and cDC2s are likely dependent on sterol metabolites for their function. Engineered LNPs are emerging as a vehicle for immunogenic delivery of mRNA vaccines, especially in the context of the severe acute respiratory syndrome coronavirus 2 pandemic (96). This study demonstrates that they might also be capable of tolerance induction in vivo, which could be highly relevant in the context of therapies for allergies or autoimmunity. The intersection of lipid metabolism with DC maturation appears to be of fundamental importance for our understanding of how our immune system distinguishes self from non-self.

MATERIALS AND METHODS

Study design

This study aimed to examine the pathways leading to homeostatic dendritic cell maturation. To this end, we used different mouse models in which we induced engulfment of fluorochrome-labeled ACs in DCs to track uptake and DC maturation by flow cytometry. In addition, we performed CITE-seq analysis to map the homeostatic dendritic cell pathway. Lipid nanoparticles were used as a source of lipids/cholesterol to load DCs and mimic engulfment of ACs. To assess the capacity to induce tolerance to dead cell-derived antigens, a cytotoxicity assay was performed. No randomization was done, and the investigator was not blinded to the mouse group allocation. Number of biological replicates is indicated as dots in the figure or as "n" in the legend. Outlier data points were removed on the basis of the robust regression and outlier removal (ROUT) method. In the bone marrow chimeric experiments, mice were omitted from the experiment when chimerism was less than 10:90.

Animal studies

Male and female WT mice C57/BL6J mice (6 to 12 weeks old) were purchased from Janvier (France) or bred at Ghent University in specific pathogen-free or germ-free conditions. *Rosa26-LSL-iDTR* [Gt(ROSA)26Sor^{tm1(HBEGF)Awai}] mice (the Jackson Laboratory, USA) were crossed to *Xcr1-Cre* (*Xcr1*^{tm1Ciphe}) mice (gift from B. Malissen, CIML, France). *Rac1*^{fl/fl} (*Rac1*^{tm1Djk}) (gift from C. Brakebusch, University of Copenhagen, Biotech Research and Innovation Centre, Denmark), *Rac2*^{-/-} (B6.129S6-*Rac2*^{tm1Mddw/J}) (V. Tybulewicz, the Francis Crick Institute, UK), *Abca1*^{fl/fl} (*Abca1*^{tm1Jp}) *Abcg1*^{fl/fl} (*Abcg1*^{tm1Tall/J}) (the Jackson Laboratory, USA), *Srebf2*^{fl/fl} (*Srebf2*^{tm1.1Jdh}) (the Jackson Laboratory, USA), and LXR α ^{fl/fl} LXR β ^{fl/fl} (gift from S. Baron, Université Clermont Auvergne, France) mice were crossed to *Itgax-Cre* (*Tg*^{Itgax-cre1-1Reiz}, CD11c-Cre) mice to generate DC-specific knockouts (Δ DC). OT-I [C57BL/6-Tg(Tcr α Tcr β)1100Mjb/Crl] mice were bought from Charles River (France) and crossed in-house to CD45.1 (B6.SJL-Ptprc^aPepc^b/BoyJ) mice, and offspring were used for experiments. ActmOVA [C57BL/6-Tg(CAG-OVAL)916Jen/J], STING^{-/-} [B6(Cg)-*Sting1*^{tm1.2Camb/J}], MyD88^{-/-} (*Myd88*^{tm1Aki}), and

TRIF^{-/-} (Ticam1^{tm1Aki}) mice were purchased from the Jackson Laboratory (USA) and bred in-house.

All mouse strains were kept on a full C57/BL6J background. All animal experiments were approved and performed in accordance with institutional guidelines for animal care of the VIB site Ghent–Ghent University Faculty of Science. To generate chimeras, CD45.1 acceptor mice were lethally irradiated (8 grays). Twelve hours later, CD45.1 acceptor mice were intravenously injected with 2×10^6 bone marrow cells of CD45.2 *Xcr1*-Cre *Rosa26*-LSL-iDTR mice or a 1:1 mix of CD45.1.2 WT and CD45.2 LXR $\alpha^{\Delta/\Delta}$ LXR $\beta^{\Delta/\Delta}$ or LXR α LXR β Δ DC bone marrow. Mice were used at least 8 weeks after reconstitution. When chimerism was less than 10:90, the mouse was omitted from the experiment.

Engulfment assay

For AC engulfment experiments, 50 million thymocytes were resuspended in 10 ml of RPMI 1640 (Thermo Fisher Scientific; 21875-059) supplemented with 10% fetal bovine serum (FBS; Bodinco), containing 10 μ M dexamethasone (Sigma-Aldrich; D2915), and incubated at 37°C in a humidified atmosphere with 5% CO₂ for 3.5 hours. When required, TopFluor-cholesterol (1 μ g/ml; Merck, 810255P) was added to the medium to load thymocytes with labeled cholesterol. Next, to allow tracking of the ACs, the cells were labeled with Cell Proliferation Dye eFluor 450 (CTV; Thermo Fisher Scientific, 65-0842-90) or Cell Proliferation Dye eFluor 670 (CTR; Thermo Fisher Scientific, 65-0840-85) according to the manufacturer's protocol. Thirty to 50 million ACs were intravenously injected. At different time points (30 min, 1 hour, 2 hours, 4 hours, 6 hours, 8 hours, 12 hours, 16 hours, 24 hours, and 48 hours) after injection, the mice were euthanized, and uptake of CTV/CTR-labeled ACs in the spleen was assessed by flow cytometry.

To investigate the uptake of beads, mice were intravenously injected with 100 μ l of 5 $\times$ diluted 0.5- μ m red fluorescent FluoSpheres (580/605) (Thermo Fisher Scientific, F8812). Two and 12 hours after injection, the mice were euthanized, and the uptake of beads was assessed by flow cytometry.

To investigate the uptake of LNPs, mice were intravenously injected with 100 μ l of Cy5-labeled eLNPs or pIC-coupled LNPs (homemade; see below). Two, 8, or 12 hours after injection, the mice were euthanized, and the uptake of LNPs was assessed by flow cytometry. When injected intramuscularly, the LNPs were 1,1-dioctadecyl-3,3,3,3-tetramethylindodicarbocyanine (DiD)-labeled instead of labeled with Cy5. The LNPs were 10-fold diluted, and 20 μ l were injected in the right quadriceps of the mouse. The draining iliac lymph node was dissected 24 hours after injection. The inguinal lymph node was used as a nondraining control.

Cytotoxicity assay

OT-I cells were isolated from CD45.1.2 OT-I mice and fluorescence-activated cell sorting (FACS)-purified on live cells, CD11c⁻, MHCII⁻, CD19⁻, CD4⁻, CD8a⁺, and CD62L⁺. A total of 0.5 million cells were adoptively transferred in acceptor mice by intravenous injection. One day later, the acceptor mice were injected with either 20 million apoptotic thymocytes derived (i) from gender-matched ActmOVA mice, (ii) from gender-matched WT mice as a negative control, or (iii) with pIC-transfected apoptotic

thymocytes from gender-matched ActmOVA mice as a positive control.

pIC was transfected in thymocytes using the Neon Transfection System 100 μ L Kit (Thermo Fisher Scientific, MPK10025) before inducing apoptosis. The cells were resuspended in buffer R at 50 million cells per ml. The Neon Transfection Device was set to a pulse voltage of 1550 V, a pulse width of 10 ms, and a pulse number of 3. One hundred microliters of the cell suspension were transfected with 10 μ g of pIC (Sigma-Aldrich, P9582).

On day 7, the acceptor mice were injected with 10 million live cells of a mixed 1:1 ratio of ActmOVA splenocytes (CTR-labeled) or WT splenocytes (CTV-labeled). One day later, OT-I cytotoxicity was measured by determining the ratio of CTV-labeled (WT) versus CTR-labeled (OVA) cells.

Statistical analysis

Statistical analyses and data representations were performed using Prism (GraphPad, La Jolla, CA). Data are presented as means $\pm$ SD or SEM as specified in the legends of the figures. Biological replicates are shown. The data distribution and variance characteristics were considered for statistical testing. In the case of comparing more than two populations, ordinary one-way analysis of variance (ANOVA) with correction for multiple testing was used for the statistical analysis. Statistical significance was defined as $P < 0.05$, and two-sided testing was used. In some graphs, the statistical significance is not shown to enable figure clarity. Asterisks in graphs represent the P value unless mentioned otherwise. Number of biological replicates is indicated as dots in the figure or as n in the legend. The investigator was not blinded to the mouse group allocation. The sample size was dependent on the availability or was determined by power analysis.

Supplementary Materials

This PDF file includes:

Supplementary Materials and Methods
Figs. S1 to S13
Tables S1 to S3
References (97–113)

Other Supplementary Material for this manuscript includes the following:

Data files S1 to S6
MDAR Reproducibility Checklist

[View/request a protocol for this paper from Bio-protocol.](#)

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LXR signaling controls homeostatic dendritic cell maturation

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Cholesterol triggers tolerogenic DCs

Whereas dendritic cells (DCs) rapidly undergo immunogenic maturation in response to pathogen exposure, the signals driving their homeostatic maturation into tolerogenic DCs that help prevent autoimmunity remain unclear. Within the mouse spleen, Bosteels *et al.* identified engulfment of apoptotic cells as a key driver of the homeostatic maturation program among type 1 conventional DCs (cDC1s). Using single cell transcriptomics, they found that pathways controlling cholesterol efflux, including the liver X receptor (LXR) pathway, were induced during homeostatic cDC1 maturation. Cholesterol uptake, either via apoptotic cells or unadjuvanted lipid nanoparticles, promoted homeostatic DC maturation. Conversely, DCs lacking LXR α and LXR β initiated an immunogenic maturation program after engulfment of apoptotic cells, demonstrating that cholesterol regulatory pathways control the decision between immunogenic and tolerogenic DC maturation. —CO

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